Q4) a: STAT 4 C03

X is an n x d matrix of 0s and 1s

. The sum of each jth column for $j = 1, \cdots, d$ is given by $\sum_{i=1}^{n} X_{ij} = n_j$ because we have n_j occurences of each level j.

. The sum of each ith row for $i = 1, \cdots, n$ is given by: $\sum_{j=1}^{d} X_{ij} = 1$ because we have a single "1" for each individual

Q4) b:

We know that $(X^T X)_{ij} = \sum_{k=1}^{n} X_{ik} X_{jk}$

We have two cases:

- if $i = j$, then

$$(X^T X)_{jj} = \sum_{k=1}^{n} X_{jk}^2 = n_j$$

- else $i \neq j$, then $X_{ik} X_{jk} = 0 \quad \forall k = 1, \dots n$

and so $(X^T X)_{\substack{ij \\ i \neq j}} = 0$

$X^T X$ is a diagonal matrix with n_j, $j = 1, \dots, d$ on the diagonals.

$$X^T X = \text{diag}(n_1, n_2, \dots, n_d)$$

and its inverse is: (suppose $n_j \neq 0 \ \forall j$)

$$(X^T X)^{-1} = \text{diag}\left(\frac{1}{n_1}, \frac{1}{n_2}, \dots, \frac{1}{n_d}\right)$$

Q4) c: STAT 4C03

We use the OLS formula :

Let $\hat{\beta} = (\beta_1, \cdots, \beta_d)$ and $y = (y_1, \cdots, y_n)$ the response variable

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

meanwhile, $(X^T y)_j = \sum_{k=1}^{n} X_{kj}\, y_k$

so finaly : $\forall\ j \in [\![1, d]\!]$

$$\hat{\beta}_j = \frac{1}{n_j} \sum_{k=1}^{n} X_{kj}\, y_k = \overline{y}_j$$

Q4) d: The residuals are given by

$$e = (e_1, \cdots, e_n) = Y - X\hat{\beta} \quad \text{and} \quad \forall\, i = 1, \cdots, n;$$

$$\boxed{e_i = y_i - \sum_{j=1}^{d} X_{ij}\, \bar{y}_j}$$

$$MSE = \frac{1}{n-d} \sum_{i=1}^{n} e_i^2$$

$$= \frac{1}{n-d} \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{d} X_{ij}\, \bar{y}_j \right)^2$$

We know that $\forall\, i = 1, \cdots, n$, $\exists!\, k(j) = 1, \cdots, d$ s.t. $X_{ik} = 1$ and 0 for all other cases $\forall j$

$$\sum_{j=1}^{d} X_{ij}\, \bar{y}_j = \bar{y}_{k(i)}$$

$$MSE = \frac{1}{n-d} \sum_{i=1}^{n} (y_i - \bar{y}_{k(i)})^2$$

let sum by level $j = 1, \cdots, d$

$$MSE = \frac{1}{n-d} \sum_{j=1}^{d} \underbrace{\sum_{i \in \text{level } j} (y_i - \bar{y}_j)^2}_{(n_j - 1) s_j^2}$$

s_j^2 is supposed to be unbiased.

finally:

$$\boxed{MSE = \frac{1}{n-d} \sum_{j=1}^{d} (n_j - 1) s_j^2}$$